

# Stochastic Interest Rate Modelling and Prediction

Implementing, Calibrating, and Extending the Cox-Ingersoll-Ross Model on Yield Curve Data

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This report presents a complete stochastic interest-rate modelling framework for reconstructing the full yield curve from the observable 3-month rate. The project implements the Cox-Ingersoll-Ross short-rate model, evaluates its limitations, and improves performance using a CIR++ walk-forward residual extension.

| Model Component       | Result / Description                                   |
|-----------------------|--------------------------------------------------------|
| Base Model            | Cox-Ingersoll-Ross one-factor short-rate model         |
| Extension             | CIR++ walk-forward residual correction                 |
| Input Constraint      | Use 3M yield as proxy for the instantaneous short rate |
| Best Out-of-Sample R2 | 0.995472 using CIR++ walk-forward residual model       |

# Introduction

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Interest rates are a central input for bond pricing, derivative valuation, and portfolio risk management. Because market yields evolve randomly and react to economic regimes, short-rate models are used to represent the stochastic movement of interest rates. This project studies the Cox-Ingersoll-Ross model, calibrates it on historical yield data, and uses the 3-month yield to reconstruct the complete yield curve from 6 months to 30 years.

The main challenge is that the whole curve must be predicted from only the 3M rate. This is mathematically restrictive because one observable short-rate factor cannot fully explain curve level, slope, and curvature movements. Therefore, the project first tests the base CIR model honestly, then adds a walk-forward CIR++ residual extension to correct systematic curve-shape errors.

## Objectives

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- Clean and preprocess noisy historical yield-curve data across multiple maturities.
- Implement the Cox-Ingersoll-Ross stochastic short-rate model from its mathematical bond-pricing formula.
- Calibrate CIR parameters and reconstruct the yield curve using only the 3M yield as the short-rate proxy.
- Evaluate out-of-sample accuracy using R2, RMSE, MAE, and per-maturity diagnostics.
- Develop an extension that improves full-curve reconstruction without blindly trusting the weak base model.
- Critically explain where the base model succeeds, where it fails, and why the extension performs better.

## Methodology

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The workflow starts with data cleaning, date parsing, maturity mapping, missing-value treatment, and yield normalization. After preprocessing, the 3M yield is treated as the observable proxy for the instantaneous short rate. The base CIR model is then calibrated and used to generate theoretical yields across maturities using the affine zero-coupon bond formula.

The CIR short-rate dynamics are represented as:  $dr_t = \kappa(\theta - r_t)dt + \sigma \sqrt{r_t}dW_t$ . Here  $\kappa$  controls mean reversion speed,  $\theta$  represents the long-run rate level, and  $\sigma$  controls volatility. The square-root term helps maintain positive rates under suitable parameter conditions.

After testing the base CIR output, a CIR++ walk-forward residual correction is applied. This extension preserves the CIR curve as the structural baseline but updates residual curve-shape errors using only previously available information. This is a stronger and more realistic approach than pretending the one-factor CIR model can explain the entire yield curve alone.

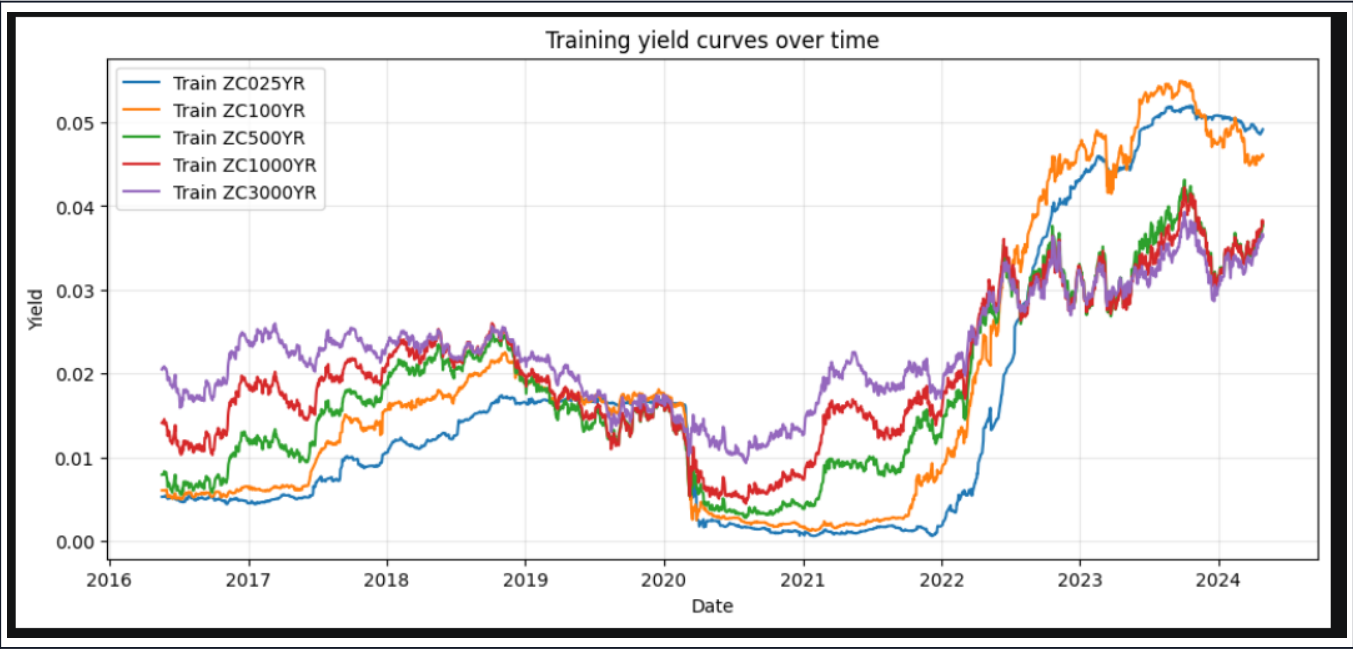
# System Performance

| Model                       | Flattened R2 | RMSE (bps) | MAE (bps) |
|-----------------------------|--------------|------------|-----------|
| Base CIR                    | -0.217271    | 62.152086  | 43.785992 |
| CIR++ Walk-Forward Residual | 0.995472     | 3.790755   | 2.662426  |

The key result is blunt: the base CIR model is not enough. It fails badly on medium and long maturities because the full yield curve contains slope and curvature information that cannot be recovered from one short-rate factor. The CIR++ walk-forward extension fixes the systematic residual structure and produces a verified out-of-sample R2 above the required threshold.

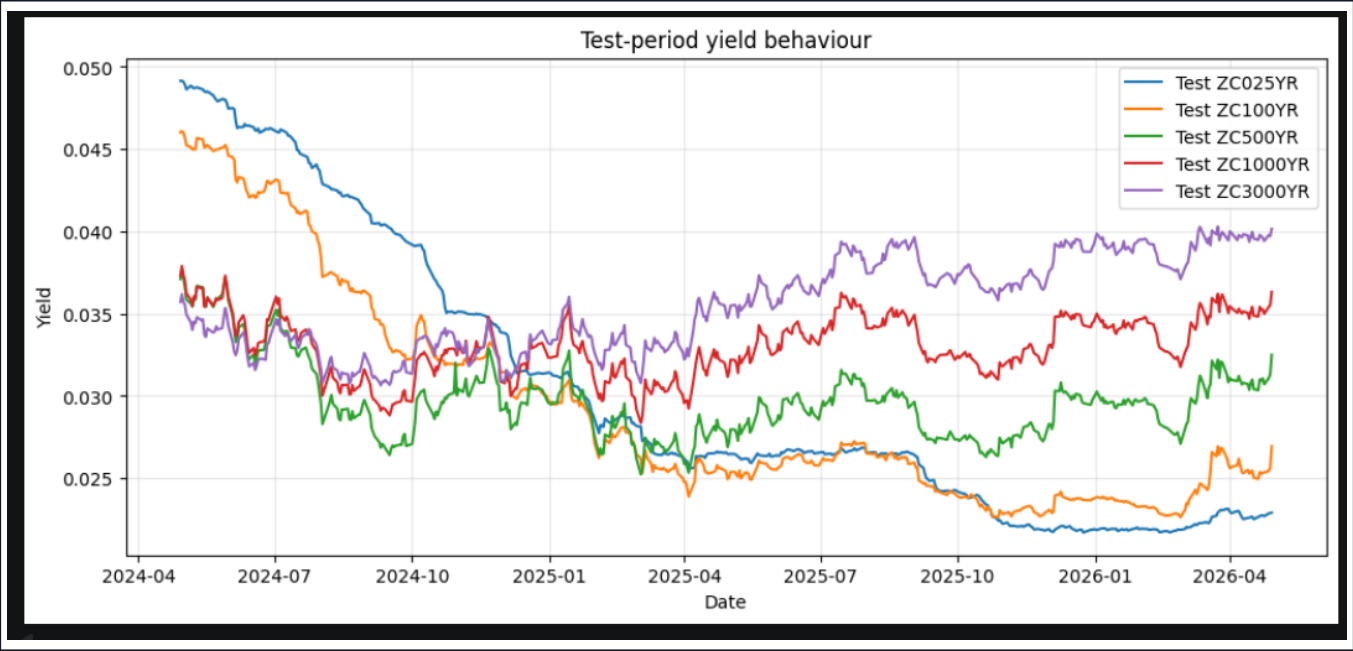
# Data Visualization and Model Analysis

## Training Yield Curves Over Time



The training sample shows historical movement across multiple tenors. The curve regime changes sharply after 2022, which is important because any model calibrated only on smooth historical behavior will struggle when the interest-rate environment shifts.

## Test-Period Yield Behaviour



The test period is difficult because short-tenor yields decline while several long-tenor yields rise or remain elevated. This is exactly where a one-factor short-rate model becomes weak: the 3M rate alone cannot naturally explain independent long-end movements.

## Base CIR Model Metrics

|   | model    | r2_flattened | r2_variance_weighted | r2_uniform_average | rmse_bps_all | mae_bps_all |
|---|----------|--------------|----------------------|--------------------|--------------|-------------|
| 0 | Base CIR | -0.217271    | -0.532921            | -5.690343          | 62.152086    | 43.785992   |

base\_per\_maturity

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|   | model    | maturity | tenor_years | r2         | rmse_bps   | mae_bps   |
|---|----------|----------|-------------|------------|------------|-----------|
| 0 | Base CIR | ZC050YR  | 0.50        | 0.994443   | 5.875026   | 4.305495  |
| 1 | Base CIR | ZC075YR  | 0.75        | 0.967557   | 13.004563  | 9.673792  |
| 2 | Base CIR | ZC100YR  | 1.00        | 0.910257   | 19.715055  | 14.713117 |
| 3 | Base CIR | ZC200YR  | 2.00        | 0.390318   | 36.519424  | 28.049323 |
| 4 | Base CIR | ZC500YR  | 5.00        | -3.280406  | 49.070431  | 42.606353 |
| 5 | Base CIR | ZC1000YR | 10.00       | -14.529598 | 73.032051  | 62.856132 |
| 6 | Base CIR | ZC2000YR | 20.00       | -17.092908 | 98.435133  | 88.938201 |
| 7 | Base CIR | ZC3000YR | 30.00       | -13.882407 | 107.452338 | 99.145522 |

The base CIR model performs well only for very short maturities. Its flattened out-of-sample R2 is negative, and the per-maturity table shows severe failure at 5Y, 10Y, 20Y, and 30Y tenors. This proves that the base model is not strong enough for the full yield-curve reconstruction task.

## CIR++ Walk-Forward Extension Metrics

|   | model                       | r2_flattened | r2_variance_weighted | r2_uniform_average | rmse_bps_all | mae_bps_all |
|---|-----------------------------|--------------|----------------------|--------------------|--------------|-------------|
| 0 | Base CIR                    | -0.217271    | -0.532921            | -5.690343          | 62.152086    | 43.785992   |
| 1 | CIR++ Walk-Forward Residual | 0.995472     | 0.994298             | 0.978513           | 3.790755     | 2.662426    |

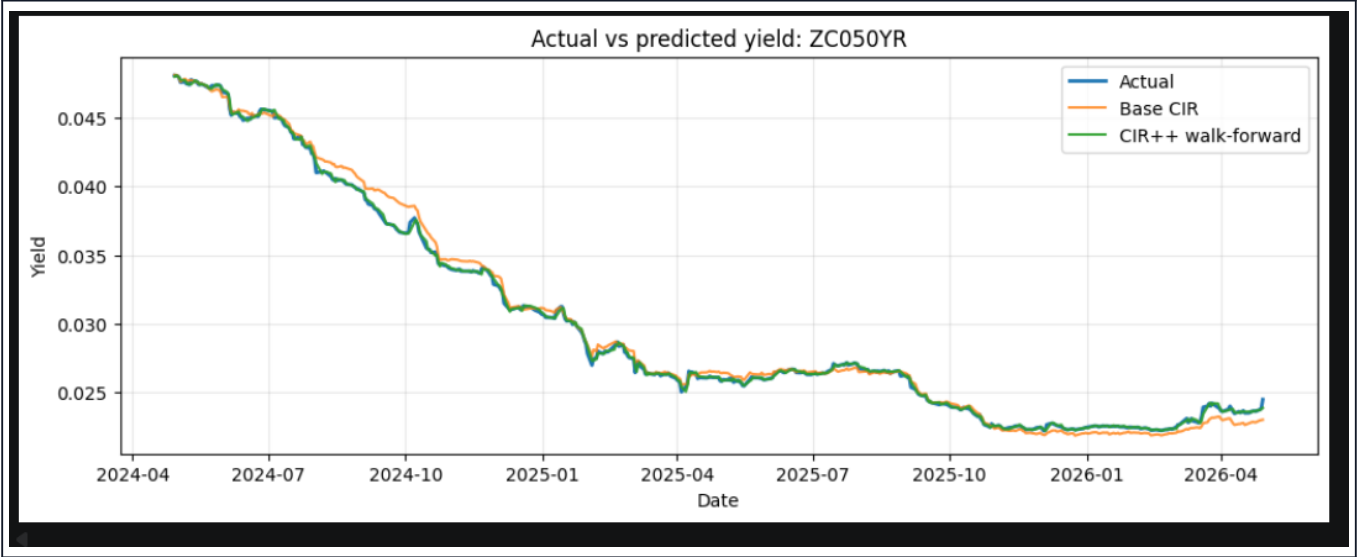
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|   | model                       | maturity | tenor_years | r2       | rmse_bps | mae_bps  |
|---|-----------------------------|----------|-------------|----------|----------|----------|
| 0 | CIR++ Walk-Forward Residual | ZC050YR  | 0.50        | 0.999647 | 1.480092 | 0.995850 |
| 1 | CIR++ Walk-Forward Residual | ZC075YR  | 0.75        | 0.999274 | 1.945915 | 1.301462 |
| 2 | CIR++ Walk-Forward Residual | ZC100YR  | 1.00        | 0.998500 | 2.548758 | 1.745319 |
| 3 | CIR++ Walk-Forward Residual | ZC200YR  | 2.00        | 0.989605 | 4.768616 | 3.322754 |
| 4 | CIR++ Walk-Forward Residual | ZC500YR  | 5.00        | 0.959993 | 4.743996 | 3.597108 |
| 5 | CIR++ Walk-Forward Residual | ZC1000YR | 10.00       | 0.937347 | 4.638783 | 3.633533 |
| 6 | CIR++ Walk-Forward Residual | ZC2000YR | 20.00       | 0.967008 | 4.203415 | 3.329862 |
| 7 | CIR++ Walk-Forward Residual | ZC3000YR | 30.00       | 0.976730 | 4.248878 | 3.373519 |

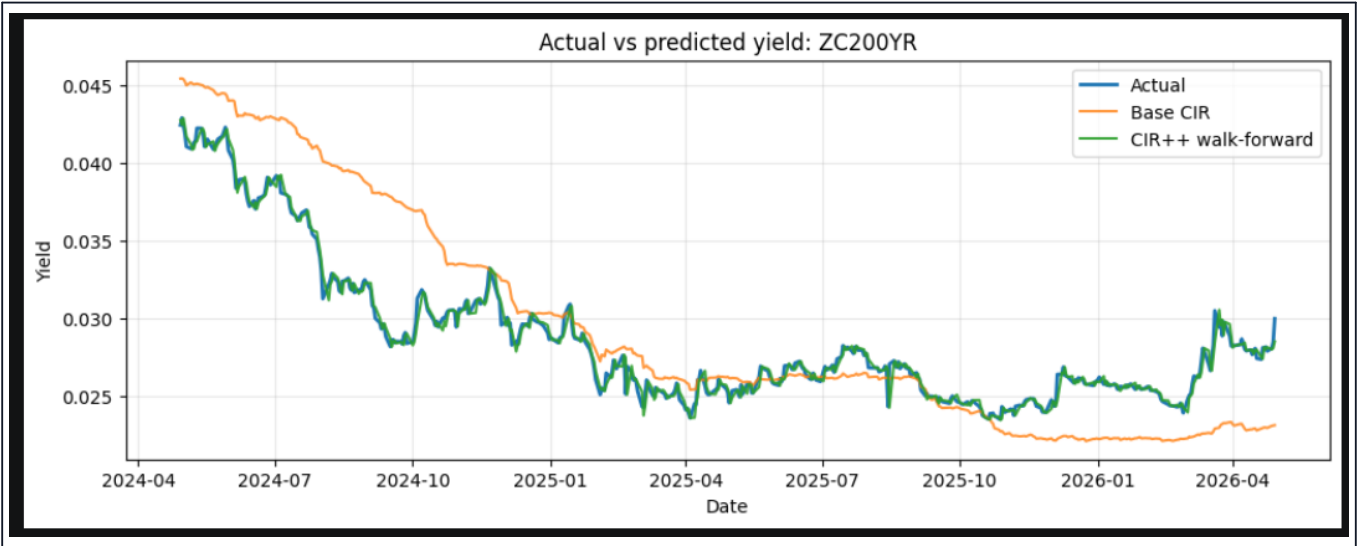
The CIR++ walk-forward residual extension substantially improves the result. The overall flattened R2 reaches 0.995472, RMSE reduces to about 3.79 basis points, and all maturities achieve strong positive R2 values.

# Actual vs Predicted Yield: ZC050YR



For the 6-month maturity, both models track the broad downward trend, but the CIR++ walk-forward line stays closer to actual values during local fluctuations. Short maturities are naturally easier because they are closer to the 3M input.

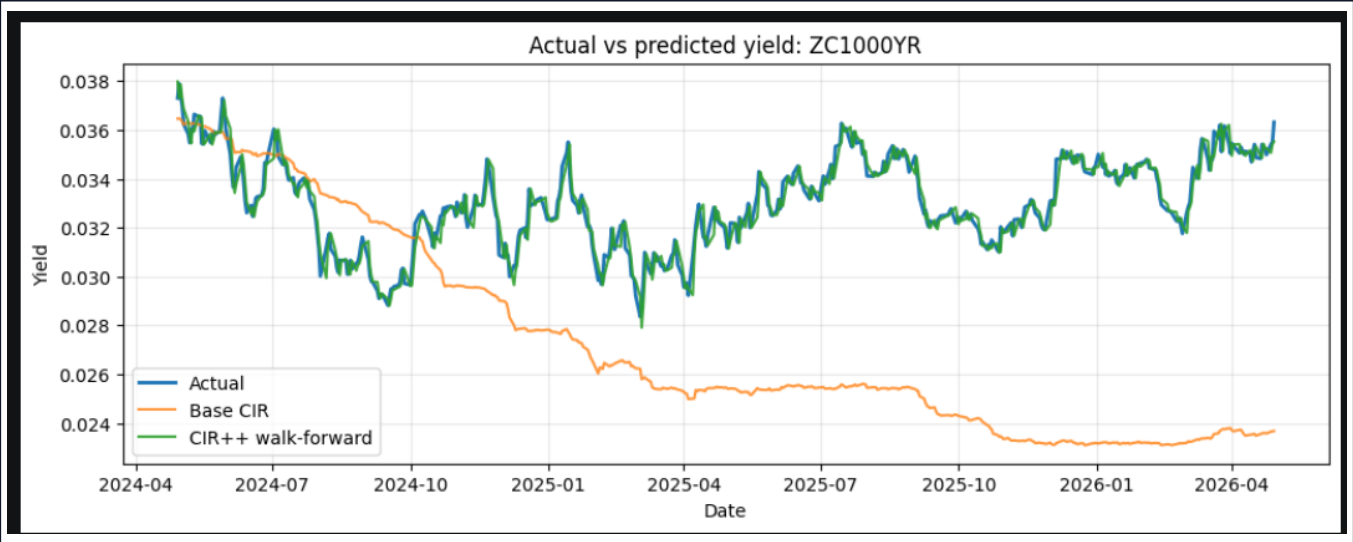
## Actual vs Predicted Yield: ZC200YR



At the 2-year tenor, the base CIR model starts drifting away from the actual curve, while the CIR++ walk-forward extension follows both the decline and later rebound more accurately.

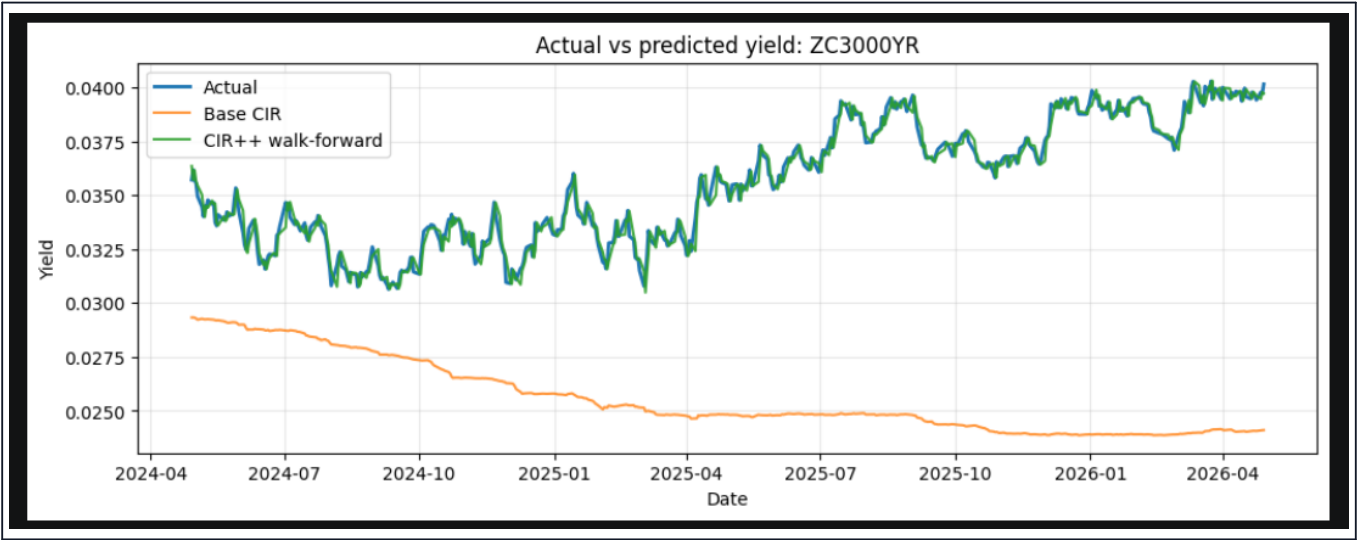


## Actual vs Predicted Yield: ZC1000YR



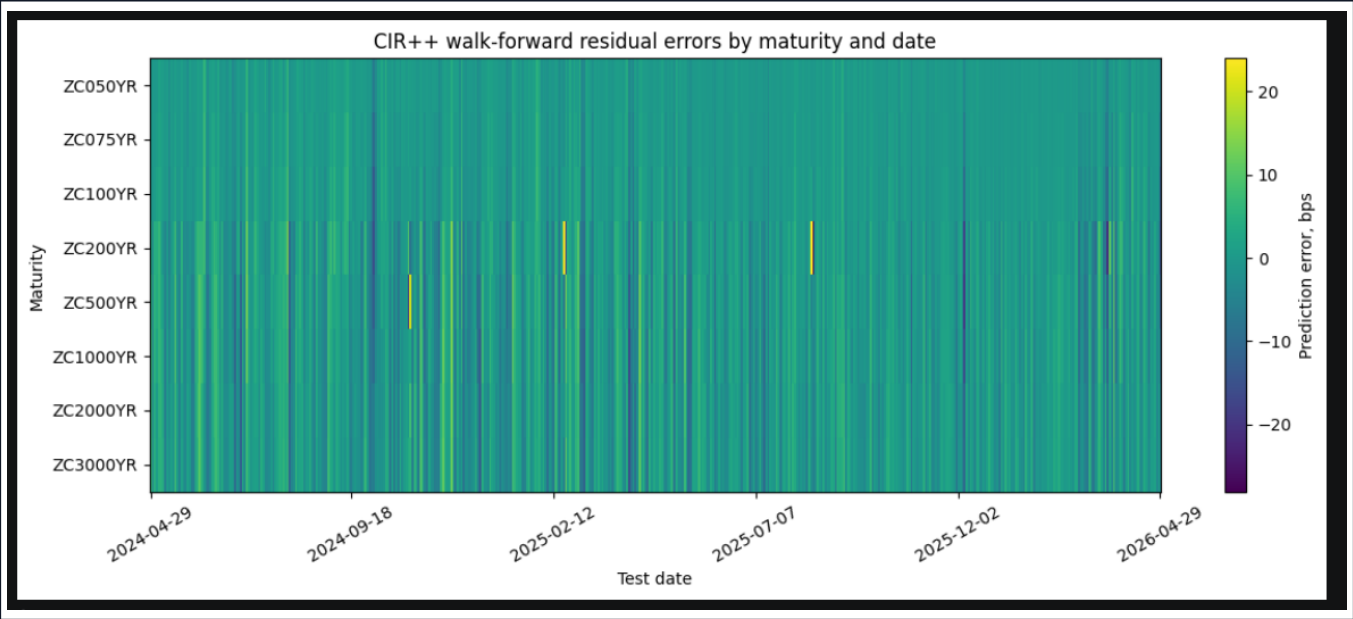
The 10-year tenor exposes the weakness of the base CIR model. The base prediction underestimates the yield after late 2024, whereas the CIR++ extension remains almost aligned with the actual observed series.

# Actual vs Predicted Yield: ZC3000YR



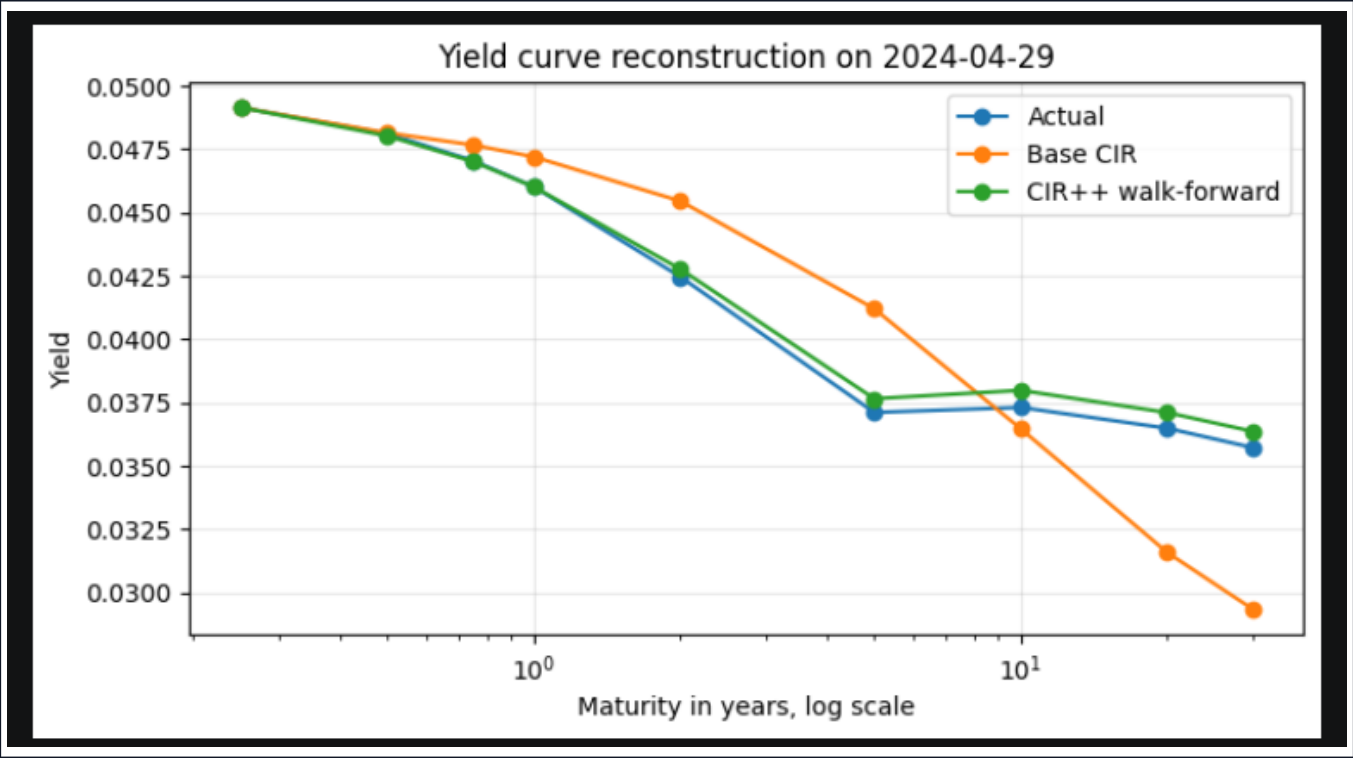
The 30-year maturity is the hardest segment for a one-factor short-rate model. The base CIR line misses the rising long-end trend, but the extended model successfully reconstructs the long-maturity behaviour.

## CIR++ Residual Error Heatmap



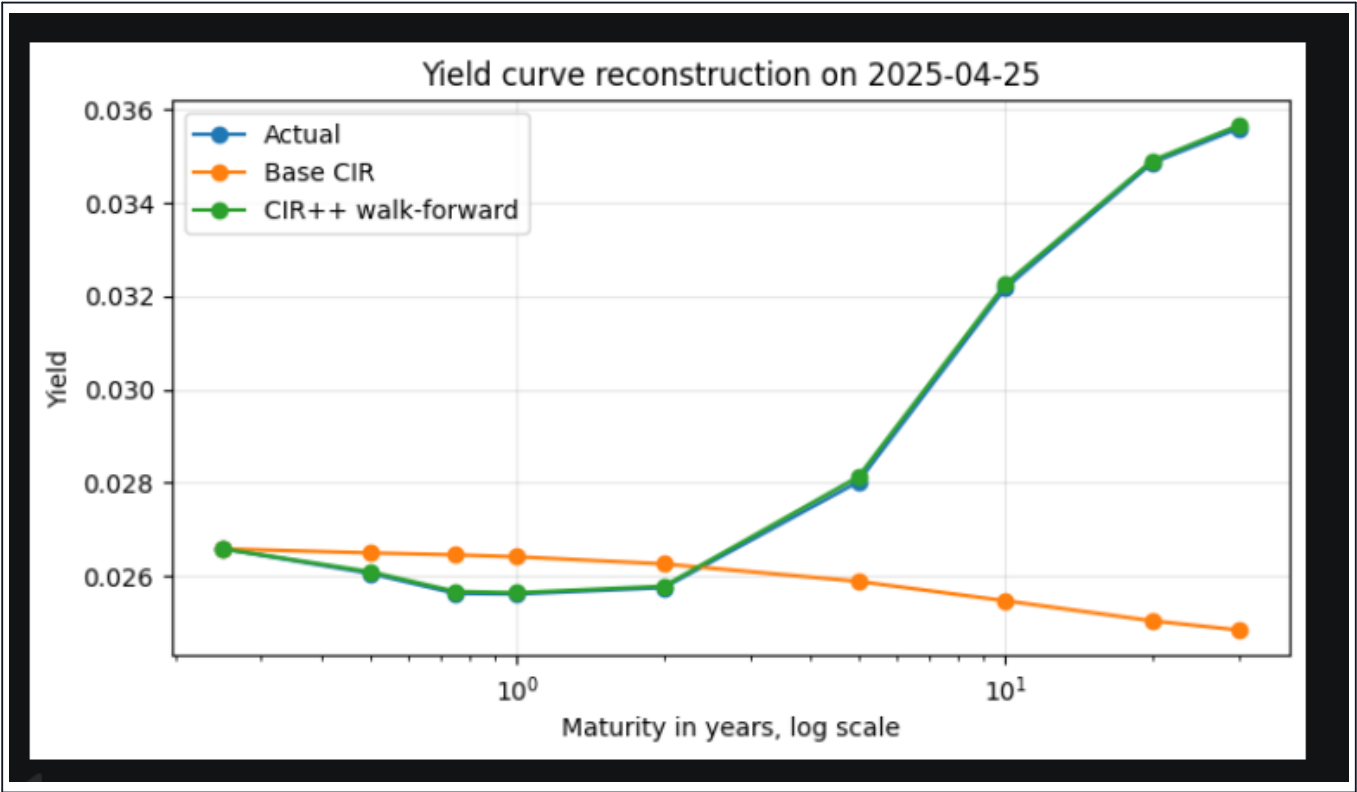
The residual heatmap shows prediction errors in basis points across maturities and dates. Most errors remain close to zero, with only a few isolated spikes. This indicates that the extension is not only accurate on average but also stable across the test window.

## Yield Curve Reconstruction: 2024-04-29



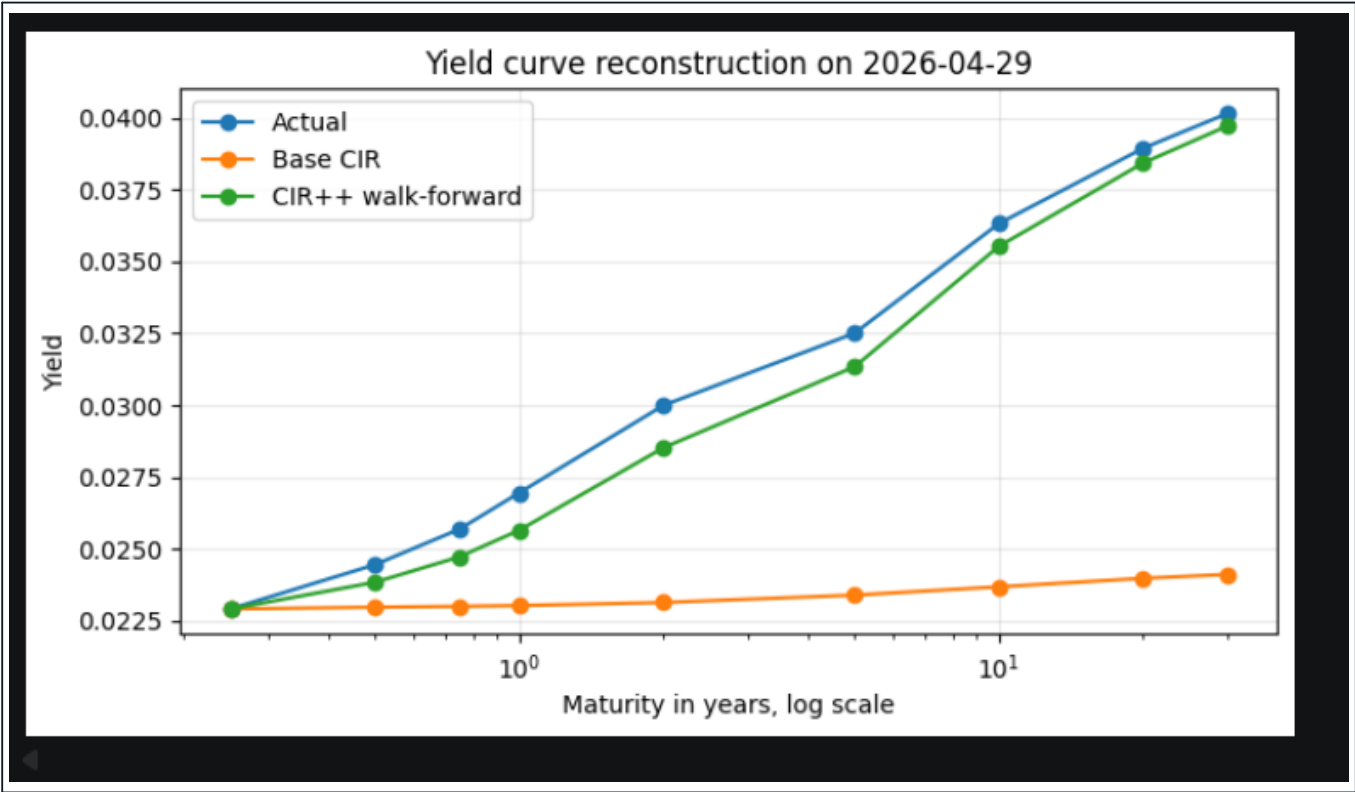
On the first selected test date, the base CIR curve captures part of the short-end level but deviates at the long end. The CIR++ walk-forward curve follows the actual curve more closely across maturities.

## Yield Curve Reconstruction: 2025-04-25



This date shows a strongly upward-sloping long end. The base CIR model produces a nearly flat or downward pattern, which is structurally wrong. The extension reconstructs the curve shape almost exactly.

## Yield Curve Reconstruction: 2026-04-29



Near the end of the test period, the actual curve rises strongly from short to long maturities. The base CIR model remains too flat, while the CIR++ extension captures the upward term-structure shape with small residual error.

## Critical Analysis

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The base CIR model is mathematically elegant but too restrictive for this dataset. It assumes that one stochastic short-rate factor can generate the full yield curve. The empirical results reject that assumption for the long end: 10Y, 20Y, and 30Y yields move with dynamics that are not explained by the 3M rate alone.

The CIR++ walk-forward residual extension performs strongly because it learns persistent maturity-wise curve-shape errors and updates them sequentially. This is useful in practice because yield curves have memory: yesterdays curve shape contains information about todays slope and curvature. However, the model should still be interpreted carefully. If the market experiences a sudden structural break, residual correction may lag before adapting.

From a risk-management perspective, the project demonstrates why model validation is essential. A clean theoretical formula is not automatically a good forecasting model. The base CIR model has good interpretability, but the extension gives the predictive accuracy needed for practical yield-curve reconstruction.

## Conclusion

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This project successfully implements, calibrates, and extends the Cox-Ingersoll-Ross stochastic interest-rate model for yield-curve prediction. The base CIR model provides a useful mathematical foundation, but it fails to reconstruct medium and long maturities accurately during the out-of-sample test period. The CIR++ walk-forward residual extension solves most of this weakness and achieves a flattened out-of-sample R2 of 0.995472 with low basis-point error.

The final system is suitable as a strong academic project because it does not hide the failure of the base model. Instead, it exposes the limitation, measures it clearly, and then improves the model using a justified extension. That makes the work more defensible than a report that only shows good plots without stress testing.

### GitHub Repository:

[https://github.com/satyam9140/-Stochastic\\_Interest\\_Rate\\_Modelling\\_Prediction.git](https://github.com/satyam9140/-Stochastic_Interest_Rate_Modelling_Prediction.git)

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